

BIPIN ADHIKARI

Machine Learning Researcher

✉ bipin.adhik@gmail.com in/bipinadhikari101 github.com/bipinadk



Professional Summary

Machine Learning Engineer with Masters in Data Science and hands-on experience in applying ML techniques (LLM, CNN, GNN, etc) to real-world systems. Combined with **MLOps experience** from Open Air Interface.

Experience

Research Engineer

Iquadrat Informatica SL

Oct 2025 - Present

Barcelona, Spain

CI and ML Developer

Open Air Interface

Feb 2024 – Sep 2024

Biot, France

- Trained ML models to **predict faulty code segments and potential errors** based on past logs and past git diffs, providing debugging assistance.
- Simplified and combined old deployment methods so different tests could run simultaneously on different servers.
- Implemented the transition of CI testing pipelines from hard-coded Python scripts to Docker Swarm and Jenkins.

Research Project Assistant (LLM / Chatbot)

Eurecom / Prof Raphael Troncy

Oct 2023 – Feb 2024

Biot, France

- Developed a LLM chatbot for the Odeuropa.eu project, handling a **dataset of 400 million** olfactory related sentences.
- Rigorously Tested and compared multiple embedding models to improve chatbot response time 250ms → 50ms.
- Implemented RAG, memory, reasoning, and multistage LLM prompting to enable LLM to think and understand the problem then find a relevant and accurate solution.

Research Projects

LLM-Based Multi-Agent System for Resource-Constrained IoT-Edge-Cloud Deployment

July 2025

- Designed and implemented a simulation of LLM-based agents (Analyzer and Planner) that translate user intents (e.g., “deploy 80 IoT sensors with low latency”) into deployment plans.
- Built structured few-shot prompts and implemented chain-of-thought reasoning for constraint-aware (CPU, RAM, bandwidth, latency) planning across heterogeneous IoT-Edge-Cloud topologies.
- **Status:** Functional early-stage prototype developed in context of a PhD application Université Côte d’Azur

Benchmarking Fine-Tuning Methods for Agentic LLMs

May 2025

- Designed a benchmarking framework for evaluating fine-tuning strategies (QLoRA, Few, RLAIIF, task-specific tuning) across diverse agentic LLM tasks like tool use, orchestration, reasoning, and multimodal understanding.
- Proposed a suite of evaluation metrics (accuracy, latency, generalizability, development cost) and defined a systematic methodology for model and task selection.
- **Status:** Extensive design and research completed; proposal prepared for PhD submission

Skills

Machine Learning: PyTorch, TensorFlow, Transformers, CNNs, GNN, Langchain, Agentic AI Systems

LLM and NLP:: Fine Tuning, Few-shot, Lora, Qlora, RAG, Chain Of Thought, BLUE, ROUGE evaluation

MLOps & DevOps: Docker Swarm, Kubernetes, CI/CD Pipelines, Git, Jenkins, AWS, MLflow

Programming & Data: Python, JavaScript/Node.js, SQL, NoSQL, VectorDBs

Languages: English (C2), French(B1), Nepali(Native), Hindi(Native)

Education

Eurecom | Institut Mines-Télécom

Master of Science in Data Science

Sep 2022 – Feb 2025

Sophia Antipolis, France

UCSI University

Bachelors of Science in Computing

Jan 2019 – Jan 2022

Kuala Lumpur, Malaysia

Chat with my AI Assistant!

adhikari.pages.dev

Ask my AI assistant about my projects, experiences, or my passions – it’s like ChatGPT, but all about me!